WHICH LOCAL BASE CHANGES OF THE FLATTENING IDEAL ARE FLAT: EXACTLY THOSE KILLED BY THE IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Base change behaviour of the flat-at-primes-over condition. Lemma \ref{lemma-base-change-flat-at-primes-over}, cited at proof lines 4476 and 4497. Gives one implication and shows that the condition passes to the quotients by powers of the maximal ideal.
- Artin-Rees: an ideal is zero in a Noetherian local ring iff it is not detected modulo every power of the maximal ideal. Algebra, Lemma \ref{algebra-lemma-intersect-powers-ideal-module-zero}, cited at proof lines 4483-4485. Replaces vanishing of the extended flattening ideal by annihilation in every Artinian quotient.
- The construction of the flattening ideal used for the flattening statement. Lemma \ref{lemma-flattening-complete-local-noetherian}, cited at proof lines 4500-4501. Reduces the claim to flatness of the module modulo the kernel of the local map at the primes over the maximal ideal.
- Flatness descends along an injective local map of Artinian rings. Algebra, Lemma \ref{algebra-lemma-descent-flatness-injective-map-artinian-rings}, cited at proof lines 4505-4506. Concludes flatness over the quotient ring from flatness over the Artinian target.

Source reference. Original lemma 0527: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6560–6560. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0526: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6559–6559. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 052W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6584–6584. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 052X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6585–6585. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00IP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 687–687. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 051L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6539–6539. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *With notation $R \to S$, M, and I and assumptions as in Lemma 0526. Consider a local homomorphism of local rings $\varphi : (R, \mathfrak{m}) \to (R', \mathfrak{m}')$ such that R' is Noetherian. Then the following are equivalent*

(1) *condition (052W) holds for $(R' \to S \otimes_R R', \mathfrak{m}', M \otimes_R R')$, and*
(2) *$\varphi(I) = 0$.*

Proof. The implication (2) $\Rightarrow$ (1) follows from Lemma 052X. Let $\varphi : R \to R'$ be as in the lemma satisfying (1). We have to show that $\varphi(I) = 0$. This is equivalent to the condition that $\varphi(I)R' = 0$. By Artin-Rees in the Noetherian local ring R' (see Algebra, Lemma 00IP) this is equivalent to the condition that $\varphi(I)R' + (\mathfrak{m}')^n = (\mathfrak{m}')^n$ for all $n > 0$. Hence this is equivalent to the condition that the composition $\varphi_n : R \to R' \to R'/(\mathfrak{m}')^n$ annihilates I for each n. Now assumption (1) for φ implies assumption (1) for φ_n by Lemma 052X. This reduces us to the case where R' is Artinian local.

Assume R' Artinian. Let $J = \mathrm{Ker}(\varphi)$. We have to show that $I \subset J$. By the construction of I in Lemma 0526 it suffices to show that $(M/JM)_{\mathfrak{q}}$ is flat over R/J for every prime $\mathfrak{q}$ of S/JS lying over $\mathfrak{m}$. As R' is Artinian, condition (1) signifies that $M \otimes_R R'$ is flat over R'. As R' is Artinian and $R/J \to R'$ is a local injective ring map, it follows that R/J is Artinian too. Hence the flatness of $M \otimes_R R' = M/JM \otimes_{R/J} R'$ over R' implies that M/JM is flat over R/J by Algebra, Lemma 051L. This concludes the proof. □